JITHIN RAJAN

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EDUCATION

University of Southern California **Aug 2025 – Dec 2027**
Master of Science in Computer Science *- GPA: 3.92/4.00* *Los Angeles, California*
Relevant Coursework: Robot Manipulation (Doctoral), Machine Learning, Applied NLP, Analysis of Algorithms, Operating Systems, Programming System Design (Java)

National Institute of Technology Tiruchirappalli **Dec 2021 – May 2025**
Bachelor of Technology in Electronics and Communication Engineering *- GPA: 3.53/4.00* *Tamil Nadu, India*
Minor in Computer Science and Engineering

TECHNICAL SKILLS

Languages: Java, Python, C++, C, JavaScript, SQL, MATLAB, HTML, Bash
Web & Backend: React, Node.js, Express.js, Flask, REST APIs, MongoDB, MySQL
Tools & Platforms: Git, Docker, Kubernetes, AWS (Lambda, API Gateway), CI/CD, Tableau
ML/Data: TensorFlow, PyTorch, OpenCV, NumPy, Pandas, Scikit-learn, Hugging Face
Certifications: Machine Learning Specialization (Stanford), Introduction to Computer Vision and Image Processing (IBM)

PUBLICATIONS

Self-Governing Assessment Network (SGAN) Based Super-Resolution for CT chest images
2023 International Conference on Computer, Electronics & Electrical Engineering & their Applications (IC2E3)

RESEARCH EXPERIENCE

USC Baum Family Maker Space – Advanced Fabrication Lab **Jan 2026 – Present**
Software & Robotics Engineer | Python, Computer Vision, Automation *Los Angeles*
- Designed and validated offline robotic simulations and toolpaths for a 6-axis Stäubli TX60 and a KUKA KR100P-2 arm using Rhino, Grasshopper, and RoboDK, catching fabrication errors before physical simulation.
- Built Python integration pipelines for real-time control of the robot, enabling online programming and closed-loop execution.
- Trained Vision Transformer-based computer vision models on live camera feeds for real-time robot perception and planning.
- Streamlined intake and fabrication workflows for 3D-printing work orders at the Makerspace across 4 manufacturing systems (Prusa MK4, Prusa HT90, Formlabs 3L, Stratasys F370), building Python scripts against Trello API to automate order processing.

Indian Institute of Technology Bombay **May 2024 – Nov 2024**
Undergraduate Research Assistant | PyTorch, MATLAB *Mumbai*
- Designed and trained ConvGRU, LSTM, and deep CNN architectures in PyTorch to estimate cerebral blood flow indices directly from optical imaging signals, replacing slower traditional computation methods.
- Built an automated signal-processing pipeline for denoising and normalizing raw optical data, cutting manual effort by over 30%.
- Benchmarked ML regressors (Random Forests, SVM, KNN) against DL-based models, achieving 15% lower prediction errors.
- Validated DL models against classical ML regressors (Random Forests, SVM, KNN), achieving a 15% reduction in prediction error.
- Worked with Dr. Hari M. Varma at the Theoretical and Experimental Biological Imaging (TEBI) Lab of the BSBE Department.

Nanyang Technological University Singapore **Jun 2023 – Sep 2023**
Research Intern | PyTorch, Sparse Matrices *Singapore*
- Investigated security vulnerabilities in gossip protocol-based federated learning systems, simulating data-poisoning attacks to measure model degradation under adversarial conditions.
- Improved model robustness, lowering attack success rate by 30% via engineered adversarial defense strategies.

Indian Institute of Technology Roorkee **May 2023 – Jul 2023**
Research Intern | Python, TensorFlow, ChimeraX *Roorkee*
- Developed a Spectral Fourier Shell Correlation (SFSC)-based alignment algorithm for GroEL cryo-electron tomography 3D reconstructions, under Dr. Jitin Singla, to improve particle-alignment accuracy prior to volume reconstruction.
- Replaced the baseline optimizer with advanced gradient descent methods, cutting computation time by 25% and improving alignment accuracy by 8%.

National Institute of Technology Tiruchirappalli **Jan 2023 – Jun 2023**
Undergraduate Research Assistant *Tiruchirappalli*
- Engineered a self governing super-resolution algorithm for CT chest imaging under Dr. S. Deivalakshmi, improving PSNR by 9.27% and SSIM by 6.17% over baseline super-resolution models.
- Co-authored and presented the resulting paper at the IC2E3 2023 conference by IEEE (NIT Uttarakhand).

PROJECTS

Academia | *Node.js, React, MongoDB, REST APIs, AWS Lambda, API Gateway, Textract*
- Contributed to a full-stack platform (Node.js/Express REST API, React/Vite frontend) helping students discover professors by research interest, integrating Typesense for fuzzy, synonym-based search that cut query latency by 10x.
- Built an event-driven paper-summarization pipeline for the platform, where an API Gateway trigger checks DynamoDB for an existing summary before queuing new requests via SQS to avoid duplicate processing.
- Built a second Lambda that extracts text from uploaded PDFs using Amazon Textract and summarizes it with the Hugging Face BART model, saving results and status to DynamoDB.

Image-to-STL Converter | *Python, FastAPI, React, PyTorch, Hugging Face Transformers*

- Built a full-stack local AI pipeline (FastAPI backend, React frontend) that converts 2D photos into watertight, 3D-printable STL meshes with zero external API cost.
- Integrated Depth-Anything-V2 for monocular depth estimation and SAM (Segment Anything) for object segmentation.
- Engineered a geometric shape classifier using mask circularity, aspect ratio, and depth-profile correlation to generate parametric meshes for recognized primitives and bas-relief heightmaps otherwise.

Co(l)de-Play | *Tezos Blockchain, Python, React*

- Built an event-ticketing platform backed by blockchain smart contracts, curbing resale price inflation by 80% and ensuring secure ownership verification and regulated ticket transfers.
- Integrated the Tezos blockchain to increase booking throughput and enable near-instant ticket processing during peak demand.
- Developed a React/Vite web application and integrated Temple Wallet for authentication and wallet management.

Digital Phenotyping | *Python, TensorFlow, Scikit-learn*

- Developed a CNN-based deep learning framework for Parkinson's disease detection from keystroke dynamics, achieving 95%+ accuracy on the Keystroke Dynamics Benchmark Dataset through feature engineering and model optimization.
- Built an end-to-end ML pipeline for preprocessing, training, and evaluation; benchmarked deep learning models against classical ML algorithms using accuracy, F1-score, and ROC-AUC metrics.

WORK EXPERIENCE

Perfios Software Solutions Pvt. Ltd. (FinTech Software Company) **Nov 2022 – Mar 2023**
Data Science and Consulting Intern | *Python, PowerBI* *Bangalore*

- Collaborated with cross-functional teams to perform an in-depth analysis of retail banking markets across 21 MENA countries, spanning market sizing, competitor dynamics, and legal compliance.
- Coordinated with product managers and business stakeholders to benchmark features of 9 SaaS products and tailor modification recommendations to align with evolving market needs.
- Delivered ML-powered loan risk and repayment models, raising risk assessment accuracy by 12% on pilot portfolios.

COURSES & CERTIFICATIONS

Machine Learning Specialisation | *Python, TensorFlow, Matplotlib* *Coursera*

- Completed a specialization of 3 courses by Dr. Andrew Ng offered by Stanford University and DeepLearning.AI.
- Implemented different models of regression and classification, neural networks, decision trees, recommender systems, unsupervised and reinforcement learning.

LEADERSHIP & ACHIEVEMENTS

Spark-A-Change Foundation **Jun 2024 – Aug 2024**
Educator *Mumbai*

- Provided academic tutoring in English and Math and increased student engagement for 100+ underprivileged children across
- Grades 1–9 through biweekly instruction.
- Organized post-class distribution of healthy meals and coordinated engaging weekend activities.

180 Degrees Consulting (NIT Trichy Chapter) **Oct 2022 – Mar 2023**
Data Science and Consulting Intern *Tiruchirappalli*

- 180 Degrees Consulting is the world's largest consultancy for non-profit organisations and social enterprises
- Restructured and executed the client acquisition process and worked on expanding the mentor network.
- Undertook one Go-To-Marketing(GTM) consulting project, conducted two case competitions and organised ConsultXpo (flagship
- annual event of the club).